

1

$$\frac{11}{20}$$

Test ①

$$\frac{22}{40}$$

$$\frac{11}{20}$$

Computer Organization MCQ

Infeepedia

By: Infee Tripathi

Assignment-#1
Computer Organization MCQ

1. Flynn's Taxonomy classifies computer architectures based on:

- (A) Number of processors and Memory units
- (B) Number of instructions and data streams
- (C) Speed of processing
- (D) More than one of the above
- (E) None of the above

2. Which of the following is NOT a category in Flynn's Taxonomy?

- (A) SISD
 - (B) MIMD
 - (C) MISD
 - (D) More than one of the above
 - (E) None of the above
- SIMD* } All are category of Flynn's taxonomy

3. An example of SIMD architecture is:

- (A) A vector processor
- (B) A multi-core processor
- (C) A pipeline processor
- (D) More than one of the above
- (E) None of the above

4. Which Flynn's taxonomy category corresponds to parallel computers?

- (A) MIMD
- (B) SIMD
- (C) SISD
- (D) More than one of the above
- (E) None of the above

5. Which category does a traditional uniprocessor belong to in Flynn's Taxonomy?

- (A) SISD
- (B) SIMD
- (C) MIMD
- (D) More than one of the above
- (E) None of the above

6. Which of the following is a characteristic of RISC architecture?

- (A) Fewer instructions
- (B) Simple instruction set
- (C) Fixed instruction length
- (D) More than one of the above
- (E) None of the above

Flynn's Taxonomy

- # of Instructions & Data streams
- SISD, SIMD, MISD, MIMD
- SIMD → Vector processor
- // computers → MIMD, SIMD
- traditional uniprocessor → SISD

Computer Organization MCQ

Infeepedia

By: Infee Tripathi

RISCCISC

Com

7. In RISC architecture, instructions are generally:

- (B) (A) Complex
(B) Simple
(C) Variable in length
(D) More than one of the above
(E) None of the above

- ① Fewer instructions
② Simple Instⁿ set
③ fixed Instⁿ length

- ① More instructions
② Variable Instⁿ length

8. CISC architecture typically has:

- (D) (A) Fewer instructions
(B) More instructions
(C) Variable instruction lengths
(D) More than one of the above
(E) None of the above

advantage of RISC over CISC

- ① lower power consumption
② faster Instⁿ execution
③ simple hardware design

9. Which of the following is a major advantage of RISC over CISC?

- (D) (A) Lower power consumption
(B) Faster instruction execution
(C) Simpler hardware design
(D) More than one of the above
(E) None of the above

CISC

- ① use large # of Addressing modes
② complex microprogramming
③ variable Instⁿ length

10. A key feature of CISC architecture is:

- (D) (A) Use of a large number of addressing modes
(B) Complex microprogramming
(C) Variable instruction length
(D) More than one of the above
(E) None of the above

11. Machine instructions typically include:

- (A, B, C) (A) Operation code (Opcode)
(B) Operands
(C) Instruction format
(D) More than one of the above
(E) None of the above

12. Which of the following are examples of arithmetic instructions?

- (A, B, C) (A) ADD
(B) SUB
(C) MUL
(D) More than one of the above
(E) None of the above

Computer Organization MCQ

Infeepedia

Instⁿ

Machine

By: Infee Tripathi

Arithmetic

Branch

13. Branch instructions are used for:

- (A) Arithmetic operations ✓
- (B) Data transfer ✓
- (C) Changing the flow of execution
- (D) More than one of the above
- (E) None of the above

① operands

② opcode

③ Instⁿ format

① Add

② sub

③ mul

changing the flow of execution

Fetch, Decode, Execute
Instⁿ cycle

14. A machine instruction is executed in which phase of the instruction cycle?

- (A) Fetch
- (B) Decode
- (C) Execute ✓
- (D) More than one of the above
- (E) None of the above

15. Which instruction type transfers data from memory to a processor register?

- (A) Arithmetic instruction
- (B) Control instruction
- (C) Data transfer instruction ✓
- (D) More than one of the above
- (E) None of the above

transfer data from memory to a processor register

Data Transfer Instruction

16. The instruction cycle consists of:

- (A) Fetch
- (B) Decode
- (C) Execute
- (D) More than one of the above ✓
- (E) None of the above

17. Which step comes first in the instruction cycle?

- (A) Decode
- (B) Fetch ✓
- (C) Execute
- (D) More than one of the above
- (E) None of the above

Fetch phase →

fetches the Instⁿ from memory

18. During the fetch phase, the CPU:

- (A) Fetches the instruction from memory ✓
- (B) Decodes the instruction
- (C) Executes the instruction
- (D) More than one of the above
- (E) None of the above

decode phase

Interpret the fetched instruction

19. The instruction cycle ends after the:

- (A) Execute phase ✓
- (B) Fetch phase

Computer Organization MCQ

Infeepedia

By: Infee Tripathi

Instruction Format

- ① structure
- ② opcode
- ③ operands & addresses

- (C) Decode phase
- (D) More than one of the above
- (E) None of the above

20. The purpose of the decode phase is to:

- (A) Interpret the fetched instruction
- (B) Load data into registers
- (C) Execute the operation
- (D) More than one of the above
- (E) None of the above

21. An instruction format defines:

- (A) The structure of an instruction
- (B) The opcode and operands
- (C) The memory addresses involved
- (D) More than one of the above
- (E) None of the above

22. The number of bits required for an instruction format is dependent on:

- (A) The number of addressing modes
- (B) The number of operands
- (C) The number of instructions
- (D) More than one of the above
- (E) None of the above

23. In a typical instruction format, the operand field contains:

- (A) Operation code
- (B) Data to be processed
- (C) Address of data or register
- (D) More than one of the above
- (E) None of the above

24. A two-address instruction format typically has:

- (A) Two operands
- (B) One operand
- (C) No operand
- (D) More than one of the above
- (E) None of the above

25. Which format is used for loading data into a register?

- (A) Zero-address instruction
- (B) One-address instruction
- (C) Two-address instruction
- (D) More than one of the above
- (E) None of the above

register ← data

26. In a single accumulator organization:

- (A) One accumulator register is used
(B) Two accumulators are used
(C) Data is loaded directly into memory
(D) More than one of the above
(E) None of the above

27. General register CPU organization:

- (A) Utilizes multiple general-purpose registers
(B) Only uses an accumulator
(C) Uses a single program counter
(D) More than one of the above
(E) None of the above

28. The stack-based CPU organization uses:

- (A) General-purpose registers
(B) A stack for storing intermediate results
(C) Multiple accumulators
(D) More than one of the above
(E) None of the above

29. Which CPU organization is most suited for high-level language support?

- (A) General register organization
(B) Single accumulator organization
(C) Stack organization
(D) More than one of the above
(E) None of the above

30. Which CPU organization typically uses two-operand instructions?

- (A) General register organization
(B) Single accumulator organization
(C) Stack-based organization
(D) More than one of the above
(E) None of the above

31. Which of the following describes a Von Neumann architecture?

- (A) Single memory for instructions and data
(B) Separate memory for instructions and data
(C) Single control unit
(D) More than one of the above
(E) None of the above

Instruction ← Memory
Data

Computer Organization MCQ

Infeepedia

By: Infec Tripathi

32. A Harvard architecture typically has:

- (A) ☒ Separate instruction and data memory
 (B) ☐ Unified instruction and data memory
 (C) ☐ Multiple control units
 (D) ☐ More than one of the above
 (E) ☐ None of the above

33. Which architecture is most commonly used in modern processors?

- (A) ☒ Von Neumann
 (B) ☐ Harvard
 (C) ☐ SISD
 (D) ☐ More than one of the above
 (E) ☐ None of the above

34. Which of the following describes a superscalar architecture?

- (A) ☐ Can issue multiple instructions per clock cycle
 (B) ☐ Executes instructions one at a time
 (C) ☒ Uses multiple pipelines
 (D) ☐ More than one of the above
 (E) ☐ None of the above

35. In a pipelined architecture, which stage overlaps with instruction fetch?

- (A) ☐ Execute
 (B) ☒ Decode
 (C) ☐ Write-back
 (D) ☐ More than one of the above
 (E) ☐ None of the above

36. A zero-address instruction format uses:

- (A) ☒ No operands
 (B) ☐ One operand
 (C) ☐ Two operands
 (D) ☐ More than one of the above
 (E) ☐ None of the above

37. The one-address instruction format is typically used in:

- (A) ☒ Stack-based CPUs
 (B) ☐ Accumulator-based CPUs
 (C) ☐ General register-based CPUs
 (D) ☐ More than one of the above
 (E) ☐ None of the above

+CPU → other format

Computer Organization MCQ

Infeepedia

By: Infee Tripathi

38. The two-address instruction format:

- (A) Has two operand addresses
- (B) Uses an accumulator
- (C) Requires a stack
- (D) More than one of the above
- (E) None of the above

Two address Instr format
→ Two operand address.
3-address I/F → uses 3 operands.
Instr format → opcode, operands, data

39. Three-address instruction format:

- (A) Has three operands
- (B) Requires a stack
- (C) Stores results in a register
- (D) More than one of the above
- (E) None of the above

40. Which of the following fields is not typically part of an instruction format?

- (A) Opcode
- (B) Operands
- (C) Program counter
- (D) More than one of the above
- (E) None of the above

Zero address Instr → no operands